at time $t$. The time path of $(1-u_t)H_t(x)$, the mass of matches producing at productivities $x$ or lower at time $t$, is given by[9]

$$\begin{aligned}\frac{d}{dt}[(1-u_t)H_t(x)] = {} & \lambda(1-u_t)[1-H_t(x)][G(x)-G(R_t)]+\theta q(\theta)u_t[G(x)-G(R_t)] \\ & -\lambda(1-u_t)H_t(x)G(R_t)-\lambda(1-u_t)H_t(x)[1-G(x)] \\ & -\delta(1-u_t)H_t(x).\end{aligned}$$

The first term accounts for the matches with productivities above $x$ that get innovations below $x$ but above $R_t$. The newly formed matches that start off with productivities no larger than $x$ are in the second term. The third term represents the matches with productivity in the interval $[R_t, x]$ that get shocks below $R_t$ and are destroyed. The fourth term accounts for those matches in the same interval that "move up" by virtue of having drawn productivities larger than $x$. The parameter $\delta < \infty$ denotes the arrival rate of an independent Poisson process that causes separations for reasons unrelated to the match-specific productivity shocks. The last term accounts for the matches in the interval $[R_t, x]$ that are destroyed for exogenous reasons.[10] In steady state,

$$\begin{aligned}H(x) &= \left[\frac{\lambda}{\delta+\lambda}+\frac{\theta q(\theta)u}{(\delta+\lambda)(1-u)}\right][G(x)-G(R)] \\ u &= \frac{\delta+\lambda G(R)}{\delta+\lambda G(R)+\theta q(\theta)[1-G(R)]}. \qquad (2)\end{aligned}$$

Using (2), the steady-state cross-sectional productivity distribution becomes

$$H(x) = \frac{G(x)-G(R)}{1-G(R)}. \qquad (3)$$

Next, I consider the problems faced by a worker and a firm. The values of unemployment and employment to a worker are denoted $U$ and $W(x)$, respectively, and solve

$$rU = b+\theta q(\theta)\int \max[W(z)-U,0]dG(z) \qquad (4)$$

$$rW(x) = w(x)+\lambda\int \max[W(z)-U,0]dG(z)-(\delta+\lambda)[W(x)-U], \qquad (5)$$

where $r$ is the discount rate, $b \geq 0$ denotes a worker's flow income while unemployed, and $w(x)$ is the wage earned by a worker employed in a match of productivity $x$.[11]

Firms can be either filled or vacant and searching. The problem of a searching firm is summarized by

$$rV = -ck+q(\theta)\int \max[J(z)-V,0]dG(z), \qquad (6)$$

where $V$ is the asset value of a vacancy and $J(x)$ the asset value of a filled job with productivity $x$. Letting $\pi(x)$ denote flow profit, $J(x)$ satisfies

$$rJ(x) = \pi(x)+\lambda\int \max[J(z)-V,0]dG(z)-(\delta+\lambda)[J(x)-V], \qquad (7)$$

9. The fact that active matches will form and continue only for productivities at least as large as $R_t$ means that $H_t(R_t)=0$. Thus, the derivation focuses on $x \geq R_t$.

10. In Section 8 I extend the analysis to the case in which the rate $\delta$ is a decreasing function of the idiosyncratic productivity parameter $x$. I will discuss the reasons why this extension may be worth exploring shortly.

11. This formulation assumes the worker suffers no disutility from work; but see Appendix B.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

where $\pi(x) = x\min(n,k) - w(x) - ck - \phi n - C(x,\phi)k$. Instantaneous profit is the residual output after the wage $w(x)$, and all other costs of production have been paid out. This formulation has three such costs: the rental on capital, $ck$; a variable cost, $\phi n$, that can be managed by varying hours; and a fixed cost $C$ per unit of capital.[12] The variable cost $\phi n$ and the fixed cost $Ck$ are introduced to allow for the possibility of labour hoarding and under-utilization of capital, two pervasive features of the data. In the presence of these costs, for some parametrizations it will be possible that at low productivity realizations, the firm may choose to keep the worker employed despite requiring that she supply zero hours. In Sections 4 and 6.2 I will show that this type of labour hoarding has interesting aggregate implications when it occurs in equilibrium.

To fix ideas, one can think of $\phi$ as the cost of electricity, for instance, with electricity usage being proportional to hours worked. Alternatively, in Appendix B I show how to modify (5) and $\pi(x)$ to get an equivalent formulation in which $\phi n$ is the worker's disutility from supplying $n$ hours to her employer. Either way, the key observation is that $\phi > 0$ is necessary for the model to display the type of labour hoarding described above. Intuitively, if the marginal product of labour, $x$, is lower than its marginal cost, $\phi$ (expressed either in terms of resources to the firm under the first interpretation or in terms of disutility to the worker under the second), then efficiency will require the match to set $n = 0$: the worker should not supply hours to the firm. Under these circumstances, whether the firm and the worker should choose to preserve the match is a different matter. One possibility is that the instances in which the firm–worker pair sets $n = 0$ are also the cases in which the pair chooses to dissolve the match. Alternatively, one could imagine that the pair may choose $n = 0$, yet decide to preserve the match, anticipating that the low realization of the idiosyncratic shock may be reversed soon. In the first scenario no hoarding is exhibited. In the latter it is. The fixed cost $C$ introduced in this section makes it possible for the second scenario to arise in an equilibrium with endogenous separations.

Notice that if $C = \phi = 0$, then the model reduces to the standard set-up of Mortensen and Pissarides (1994). For this special case, all the results on how labour-market policies affect the level of TFP still apply even though the equilibrium does not exhibit hoarding. In this sense, neither of these costs is essential. However, in the presence of the variable cost $\phi$ and the fixed cost $C$, the model is also able to deliver equilibrium hoarding, and as it turns out, this feature has interesting implications for how the aggregate production function looks.

Since the fixed cost $C$ is perhaps the only non-standard element of the model, several remarks are in order. First, at a technical level, incorporating a fixed-cost specification that is decreasing in the shock $x$ is introduced in this section as a simple device to avoid a "flat spot" in flow profit that would otherwise carry over to the value functions.[13] Second, in Section 8 I drop the fixed cost and redo the analysis in a version of the model in which the draws of the idiosyncratic shock are correlated over time. In another extension explored in Section 8, I again drop the fixed cost and show that the key insights also apply in a version of the model in which the destruction rate $\delta$ is decreasing in the productivity shock. I wish to emphasize that these alternative specifications seem natural (perhaps even more so than the usual benchmark in which the shocks hitting the match are independent of its state) and are able to deliver hoarding without the fixed cost $C$.[14] Given all this, and for expositional purposes, I will—for the time being—use a particularly convenient specification for the fixed cost, namely, $C(x,\phi) = \max(\phi - x, 0)$.[15]

12. The cost $C$ is fixed in the sense that the firm can avoid it only by shutting down, but—as the notation indicates—it may depend on the realization of the idiosyncratic shock and possibly also on the parameter $\phi$. More on this is given below.

13. See Appendix B, in particular, the discussion around equation (47), for details.

14. Flow profit has a flat spot in both of these alternative formulations, but the serially correlated shocks ensure that the value functions do not inherit this flat spot.

15. This formulation is convenient because it will imply that the flow profit is affine and strictly increasing in $x$. Since—even accepting the presence of a fixed cost—this particular formulation may seem somewhat contrived, in

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

## 3. EQUILIBRIUM

I follow the bulk of the search literature by letting $\beta \in [0,1)$ and assuming the instantaneous wage $w(x)$ and labour supply $n$ solve max $[W(x)-U]^{\beta}[J(x)-V]^{1-\beta}$ at all times. The optimal choice of hours is

$$n(x) = \begin{cases} k & \text{if } \phi < x \\ 0 & \text{if } x \leq \phi, \end{cases} \tag{8}$$

and substituting it into the flow profit function gives $\pi(x) = (x-\phi-c)k - w(x)$ for all $x$. The first-order condition for the instantaneous wage, $w(x)$, is

$$(1-\beta)[W(x)-U] = \beta[J(x)-V]. \tag{9}$$

Letting $S(x) = J(x) + W(x) - U - V$ denote the surplus from a match, notice that (9) implies $J(x) = (1-\beta)S(x)$ and $W(x) - U = \beta S(x)$. These, together with (4), (5), and (7), imply

$$(r+\delta+\lambda)S(x) = (x-\phi-c)k - rU + \lambda \int \max[S(z),0]dG(z),$$

where

$$rU = b + \frac{\beta}{1-\beta}kc\theta. \tag{10}$$

In deriving (10) I have already acknowledged that free entry of firms will cause $rV = 0$ in equilibrium. Since $S'(x) = \frac{k}{r+\delta+\lambda} > 0$, there exists a unique $R$ such that $S(x) > 0$ iff $x > R$. Hence, a firm–worker pair destroys an existing match (and chooses not to form a new match) if it draws a productivity $x < R$. With this reservation strategy, the surplus can be written as

$$(r+\delta+\lambda)S(x) = (x-\phi-c)k - rU + \lambda \int_R S(z)dG(z). \tag{11}$$

The value functions and (9) can be combined to obtain the instantaneous wages and profit:

$$w(x) = \beta(x-\phi-c)k + (1-\beta)rU \tag{12}$$

$$\pi(x) = (1-\beta)[(x-\phi-c)k - rU]. \tag{13}$$

Intuitively, the wage is a weighted average of output (net of the rental on capital and the variable and fixed costs) and the worker's reservation wage.

Next, I characterize the job-creation and job-destruction decisions as summarized by $\theta$ and $R$, respectively. Evaluating (11) at $x = R$ yields $\lambda \int_R S(z)dG(z) = rU - (R-\phi-c)k$. Notice that since the expected capital gain on the L.H.S. is positive, at $x = R$ net output is smaller than the worker's reservation wage. Thus, (12) and (13) imply that $w(R) < rU$ and $\pi(R) < 0$: workers and firms sometimes tolerate instantaneous pay-offs below those they could get by separating, in anticipation of future productivity improvements.[16] Substituting this simpler expression for the expected capital gain term into (11) gives

$$S(x) = \frac{x-R}{r+\delta+\lambda}k. \tag{14}$$

Appendix B I redo the whole analysis with a more general specification for the fixed cost and show that $\frac{\partial C(x,\phi)}{\partial x} < 0$ is all that is needed. But again, see Section 8 for alternative specifications that do not rely on firms having to bear *any* fixed cost $C$.

16. This feature of the model is a consequence of the costly and time-consuming meeting process, as noted by Mortensen and Pissarides (1994).

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

Evaluating (11) at $x = R$ and using (14) to substitute $S(\cdot)$ yields what is usually referred to as the job-destruction condition:

$$R - \phi - c - \left(\frac{b}{k} + \frac{\beta}{1-\beta}c\theta\right) + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0. \tag{15}$$

As is standard, the destruction decision is independent of scale if $b$ is. The natural interpretation of $b$ is that it is unemployment insurance income. Along these lines, if one lets $b = \tau_b E_G[w(x) \mid x \geq R]$, where $\tau_b \in [0, 1)$ is the replacement rate, then $b = \hat{b}k$, with

$$\hat{b} = \frac{\tau_b \beta[\tilde{x}(R) - \phi - c + c\theta]}{1-(1-\beta)\tau_b},$$

and $\tilde{x}(R) \equiv E_G[x \mid x \geq R] = [1 - G(R)]^{-1}\int_R x dG(x)$. Under this specification, $b$ is linear in $k$ so (15) is independent of $k$ and becomes

$$R - \frac{\tau_b\beta\tilde{x}(R)}{1-(1-\beta)\tau_b} - \frac{(1-\tau_b)(\phi+c)}{1-(1-\beta)\tau_b} - \frac{\beta c\theta}{(1-\beta)[1-(1-\beta)\tau_b]} + \frac{\lambda}{r+\delta+\lambda}\int_R (x-R)dG(x) = 0.$$

In what follows I will always abstract from scale effects caused by unemployment income $b$ by assuming it is a fraction of the average going wage.

Substituting the equilibrium condition $rV = 0$ in (6) implies $(1-\beta)\int_R S(x)dG(x) = \frac{ck}{q(\theta)}$. That is, the expected profit from a filled job equals the expected hiring cost in an equilibrium with free entry. Using (14) to substitute $S(\cdot)$ out of this expression yields what is often referred to as the job-creation condition:

$$\frac{1-\beta}{r+\delta+\lambda}\int_R (x-R)dG(x) - \frac{c}{q(\theta)} = 0. \tag{16}$$

The job-creation and job-destruction conditions jointly determine $R$ and $\theta$, and under the maintained assumptions, they are independent of scale, $k$.[17] For given $c$ and $\phi$, an equilibrium is a vector $[\theta, R, H, U, w, u, K]$ such that $(\theta, R)$ jointly solve (15) and (16); given $(\theta, R)$, $H$ satisfies (3); $U$ is given by (10); $w$ is given by (12); and $u$ is given by (2). In addition, the market for capital services should clear, so the aggregate supply of capital $K$ must satisfy $K = [1-(1-\theta)u]k$, where the R.H.S. is the total demand for capital (coming from both matched and unmatched firms). Note that in parametrizations that result in $R < \phi$, the capital and workers in matches with realizations in $[R, \phi)$ remain employed but are not engaged in production. The firms in these states have excess capacity and hoard labour. The following section provides a sharper characterization of aggregate outcomes for a particular distribution of idiosyncratic shocks.

## 4. AGGREGATION

Let $K_e$ denote the capital in place at all the firms with filled jobs, that is,

$$K_e = \frac{1-u}{1-(1-\theta)u}K. \tag{17}$$

17. See Lemma 1 in Appendix B for conditions under which the pair $(\theta, R)$ that solves (15) and (16) exists and is unique.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms